

ABERDARE TECHNICAL COLLEGE

INFORMATION COMMUNICATION

TECHNOLOGY

ELEMENT: DESIGN COMPUTER NETWORK

UNIT CODE: IT/CU/ICTA/CR/02/5/MA

UNIT NAME: NETWORK DESIGN & MANAGEMENT

LEVEL 5, MODULE 3

TERM 3

TOTAL MARKS: 50 MARKS

Name.....Date.....

Sign.....Cdacc regNo.....

Instructions:

Answer all questions.

- 1.** Define a computer network and state two benefits of computer networking. **(4 marks)**
- 2.** State and explain four types of computer networks based on geographical coverage. **(8 marks)**
- 3.** Explain the following network topologies:
 - a) Star
 - b) Ring
 - c) Mesh
 - d) Hybrid
 - e) Point-to-point **(10 marks)**
- 4.** State four components of a computer network and explain the function of each. **(8 marks)**
- 5.** State three examples of network transmission media and give one characteristic of each. **(6 marks)**
- 6.** Explain the following stages involved in collecting and handling computer network user requirements:
 - a) User requirements identification

b) User requirements analysis

c) User requirements documentation (**6 marks**)

7. Differentiate between physical network design and logical network design. (**4 marks**)

8. Explain the difference between a **greenfield** network site and a **brownfield** network site. (**2 marks**)

9. State two fundamental goals of computer network design. (**2 marks**)